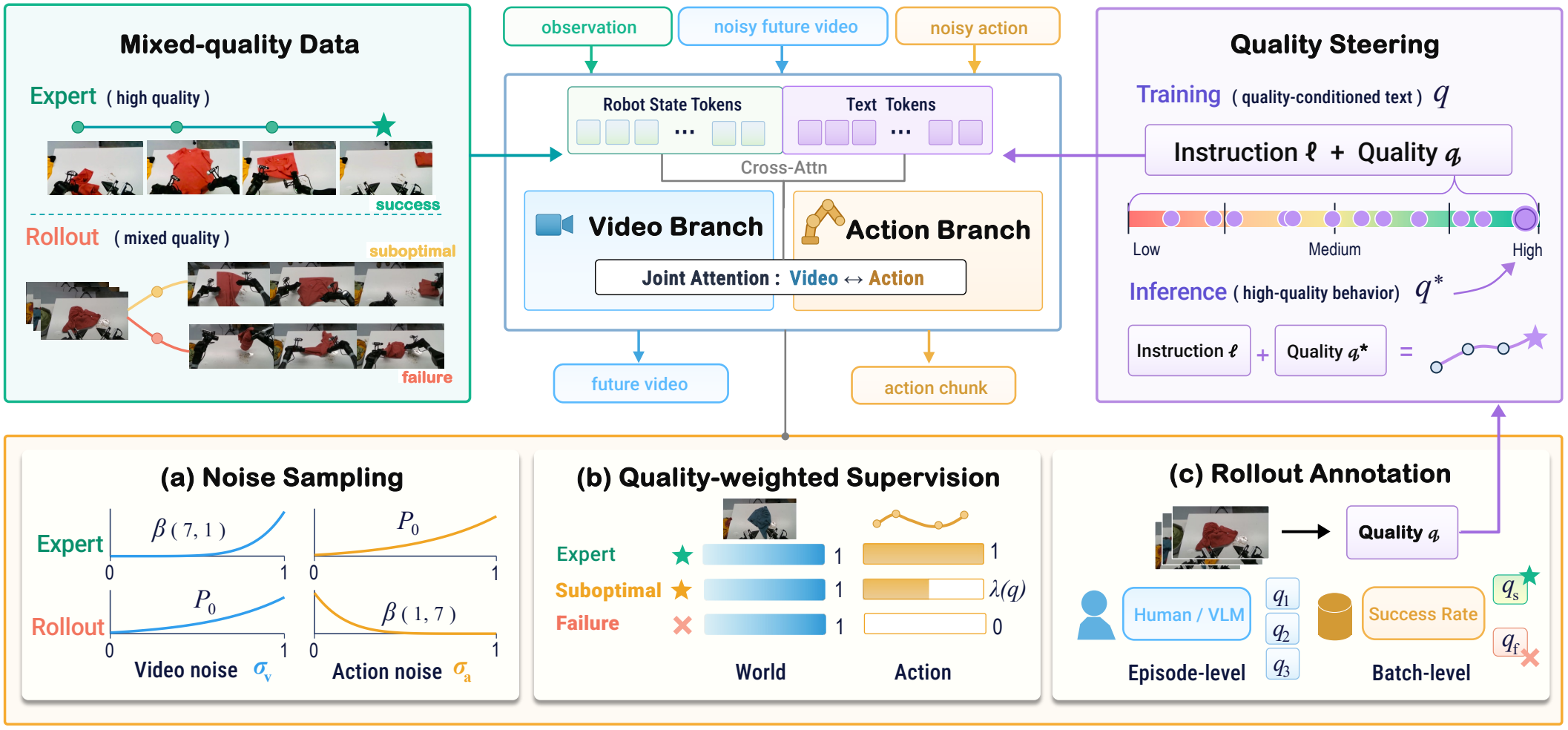

Mixed-quality Data
Expert (high quality)
success
Rollout (mixed quality)
suboptimal
failure
observation
noisy future video
noisy action
Robot State Tokens
Text Tokens
Cross-Attn
Video Branch
Action Branch
Joint Attention : Video ↔ Action
future video
action chunk
Quality Steering
Training (quality-conditioned text) q
Instruction ℓ + Quality q
Low
Medium
High
Inference (high-quality behavior) q*
Instruction ℓ
+
Quality q*
=
(a) Noise Sampling
Expert
β (7, 1)
P0
Rollout
P0
β (1, 7)
0
1
Video noise σv
Action noise σa
(b) Quality-weighted Supervision
Expert
Suboptimal
Failure
1
λ(q)
0
World
Action
(c) Rollout Annotation
Quality q
Human / VLM
Success Rate
q1
q2
q3
qs
qf
Episode-level
Batch-level